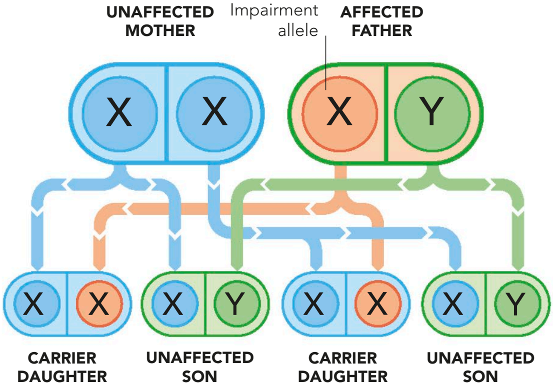
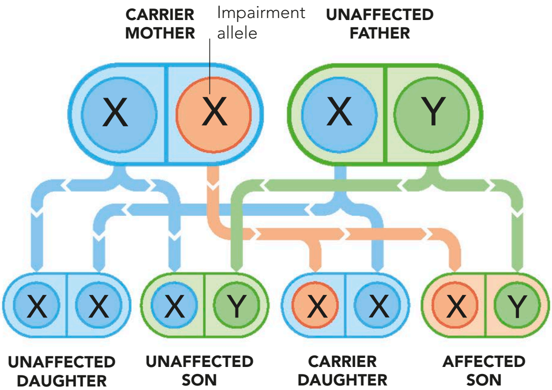
SEX-LINKED INHERITANCE

The pattern of inheritance changes when alleles for a body feature are carried on the sex chromosomes. If an allele on a man's X chromosome does not have its equal on the Y chromosome, or vice versa, only one allele can determine the feature. For example, the problem allele for colour-impaired vision is on the X chromosome.



COLOUR-BLIND FATHER AND UNAFFECTED MOTHER

Sex chromosomes combine in four possible ways, governed by chance. Here, any daughter will inherit the colour-impairment allele, and will be a carrier, but she also has the normal allele on her other X chromosome, to give normal vision. No sons can be affected, nor can they be carriers.



CARRIER MOTHER AND UNAFFECTED FATHER

The four possible combinations give a one-in-four-chance each for unaffected sons and daughters. There is also a one-in-four chance a daughter is a carrier, or that a son inherits the colour-impairment allele. He has no second X chromosome and therefore no normal allele, so the result is impaired colour vision.

MULTIPLE-GENE INHERITANCE

Some body traits follow clear single-gene inheritance patterns. However, the situation becomes more complex in two ways. First, there may not be only two alleles of a gene with a simple dominant–recessive interaction between them. There may be three alleles or more in existence in the general population, although each person can have only two of them. An example is the blood group system, with alleles for A, B, and O. Second, a trait may be influenced by more than one gene. These two situations mean that a trait can be governed by multiple genes, and for each of these genes, by multiple alleles of the gene – added to which, the genes may interact in different ways, according to which alleles are present in each of them. In such cases, the numbers of possible combinations multiply, consequently making multi-gene inheritance exceptionally difficult to unravel.